

TONGLIAO, China

WMO: 541350

Lat: 43.5954N

Lon: 122.2603E

Elev: 180

StdP: 99.18

Time Zone: 8.00 (E08)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-22.0	-19.9	-30.6	0.2	-17.9	-28.5	0.3	-16.3	10.2	-8.5	8.4	-8.9	2.2	290	0.519

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	9.1	33.1	21.2	31.6	21.1	30.3	20.5	25.0	29.2	24.1	28.1	23.1	27.2	3.9	200

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
23.8	19.0	27.5	22.8	17.9	26.8	21.9	16.9	25.8	77.7	29.2	73.9	27.9	70.1	27.0	28.2

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max		
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
9.8	8.2	7.0	-24.2	36.0	3.0	2.3	-26.4	37.7	-28.2	39.0	-29.9	40.3	-32.1	41.9		
			DB	WB	DB	WB	DB	WB	DB	WB	DB	WB	DB	WB		
			-24.6	26.3	2.9	1.3	-26.7	27.3	-28.4	28.0	-30.0	28.7	-32.1	29.7		

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
Temperatures, Degree-Days and Degree-Hours	DBAvg	7.8	-12.2	-7.3	0.3	9.8	17.2	22.3	24.9	23.2	17.5	8.9	-1.8	-10.0
	DBStd	13.67	4.23	4.96	5.33	4.75	4.03	3.13	2.29	2.57	3.53	4.72	5.72	4.43
	HDD10.0	2591	687	484	303	61	2	0	0	0	1	77	355	621
	HDD18.3	4388	946	717	558	259	70	5	0	2	54	293	604	879
	CDD10.0	1792	0	0	3	54	224	370	461	410	227	42	1	0
	CDD18.3	548	0	0	0	2	34	125	202	153	31	1	0	0
	CDH23.3	4853	0	0	1	38	392	1156	1772	1203	278	13	0	0
	CDH26.7	1657	0	0	0	7	130	449	651	368	52	1	0	0
Wind	WSAvg	3.2	2.9	3.2	3.8	4.1	3.9	3.2	2.9	2.6	2.9	3.2	3.1	2.9
Precipitation	PrecAvg	373	1	3	6	15	29	68	114	81	33	18	5	2
	PrecMax	549	6	11	40	83	69	127	297	225	82	86	17	23
	PrecMin	200	0	0	0	0	4	4	48	3	1	1	0	0
	PrecStd	77	2	3	8	17	18	35	54	54	21	20	5	5
Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	2.2	8.9	19.7	27.3	33.2	35.6	35.1	33.5	30.2	25.2	16.0	5.6
		MCWB	-2.1	1.4	8.9	13.1	16.2	19.8	23.8	22.7	19.6	14.8	7.9	1.5
	2%	DB	-1.1	5.8	14.8	23.9	30.1	33.0	33.1	31.5	28.2	21.9	12.2	1.4
		MCWB	-4.7	0.0	5.7	11.5	16.1	19.4	23.0	22.4	18.3	13.2	6.3	-2.3
	5%	DB	-3.2	3.5	11.7	21.2	27.5	31.1	31.6	30.1	26.4	19.0	9.3	-1.0
		MCWB	-6.3	-1.3	3.8	10.3	15.0	18.9	22.6	21.6	17.0	11.4	4.3	-4.2
	10%	DB	-5.2	1.0	9.0	18.5	25.1	29.3	30.2	28.7	24.7	16.7	6.6	-3.3
		MCWB	-7.8	-3.1	2.2	8.9	13.7	18.5	22.0	21.1	16.2	10.2	2.1	-5.9
Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	-1.8	2.6	9.3	14.7	19.0	23.1	26.8	25.9	22.4	16.9	9.6	1.7
		MCDB	2.0	7.6	18.1	23.8	28.4	30.0	32.5	30.5	27.4	20.6	13.0	5.3
	2%	WB	-4.6	0.4	6.2	12.8	17.3	21.9	25.5	24.8	20.5	14.8	6.8	-2.1
		MCDB	-1.4	5.5	13.7	21.4	26.4	28.2	30.3	28.7	25.1	19.8	12.0	1.1
	5%	WB	-6.2	-1.2	4.3	11.1	16.2	21.0	24.5	23.9	19.3	13.0	4.4	-4.2
		MCDB	-3.2	3.4	11.1	19.4	24.5	26.9	28.7	27.9	23.7	17.6	9.1	-1.1
	10%	WB	-7.8	-3.0	2.6	9.6	15.0	20.1	23.7	23.0	18.0	11.1	2.2	-5.9
		MCDB	-5.3	1.0	8.6	17.6	23.2	25.8	27.6	26.6	22.6	15.9	6.4	-3.3
Mean Daily Temperature Range	5% DB	MDBR	9.8	10.6	11.0	11.8	11.7	10.5	9.1	9.5	11.9	11.2	9.8	9.4
		MCDBR	12.5	13.4	15.0	16.2	15.7	13.9	11.6	11.4	14.1	14.6	13.2	11.8
		MCWBR	9.8	9.3	8.7	8.0	6.3	5.0	4.2	4.4	6.0	7.4	8.5	9.0
	5% WB	MCDBR	12.4	13.2	14.2	14.1	13.2	10.5	9.4	9.0	11.0	12.3	12.9	11.7
		MCWBR	9.8	9.4	8.5	7.5	6.0	4.5	4.4	4.4	5.6	7.5	8.8	9.0
Clear-Sky Solar Irradiance	taub	0.305	0.342	0.401	0.503	0.576	0.643	0.600	0.538	0.471	0.438	0.357	0.316	
	taud	2.190	2.114	1.995	1.786	1.683	1.637	1.776	1.903	1.982	1.959	2.094	2.142	
	Ebn at Noon	803	831	822	763	720	673	698	728	744	702	717	739	
	Edh at Noon	93	120	155	208	237	249	215	183	156	140	102	89	
All-Sky Solar Radiation	RadAvg	2.28	3.28	4.46	5.44	5.99	5.67	5.50	5.26	4.62	3.33	2.28	1.89	
	RadStd	0.17	0.21	0.26	0.36	0.40	0.46	0.44	0.28	0.22	0.22	0.16	0.15	

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46)	Station Only	N/A	N/A	-0.97	N/A	N/A	N/A	N/A	N/A	+67	N/A
(47)	Regional (5 neighbors)	N/A	N/A	-0.97	N/A	N/A	N/A	N/A	N/A	+49	N/A

Nomenclature: See separate page